Research brief: attack-resilient embedded safety

Does core pinning isolate the safety function on a single-chip Wi-Fi SoC?

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The question

Consumer intrusion alarms are built almost entirely on single-chip Wi-Fi SoCs of the ESP32 class, where the Wi-Fi driver, the TCP/IP stack, the MQTT client and the safety application all run on the same die, under one RTOS scheduler, sharing one interrupt controller. The standard advice is to pin the safety task to its own core—but that advice comes from a literature studying much larger machines, in which the attacked network interface is *physically separate* from the protected core. Does it still hold when the attacked resource is the chip's own radio driver?

The gap, from the current literature

Fourteen prior works were surveyed and individually verified against publisher records (`docs/RELATED_WORK.md`). The gap is narrow and real:

- **Timing isolation under attack** is established, but only on platforms that can separate the attacked interface from the protected core: Linux-class industrial boards (Fischer, 2020), x86 with PREEMPT-RT and dedicated NIC queues (Meszlényi, 2025), Cortex-A53 with a hypervisor and an FPGA memory scheduler (Hoornaert, *Real-Time Systems*, 2024). None of these mechanisms—MMU, hypervisor, counter-based regulation, hardware queues—exist on a 240 MHz MCU.
- **Radio-layer attacks on this chip family** are cheap and demonstrated (Kremser and Graffi, *Sensors*, 2026; Cayre *et al.*, WOOT'23), and deauthentication resilience has been quantified (Carbajal, 2026)—but the dependent variable is always whether the *link* drops, never whether the device's *application* still meets its deadline.
- **ESP32 security work** measures MQTT-over-TLS cost or on-device detection accuracy. No one treats a safety deadline as the dependent variable.

No published work measures a safety deadline on a device whose own radio driver is the attacked resource. That is the contribution on offer.

What is already established

All of the following is committed, seeded and reproducible by a single script; CI re-runs a reduced version of every one of them on each push.

Evidence	Scale	Result
TLA+ model checking (TLC)	2 models, 25 + 660 states	All properties hold; both mutation-tested
Adversarial campaign	6,000 seeded trials	0 attacker disarms, 0 invariant violations
Fuzzing (ASan + UBSan)	3,000,000 iterations	0 memory errors, 0 accepted inputs

The second TLA+ model is the methodological point worth discussing: the network task is given *no fairness assumption*—free to stall unboundedly, flood, fail to publish or crash—and the safety step is proved to remain enabled regardless. That is a formal statement of the isolation claim, independent of any particular workload.

What is open, and why it cannot be simulated

Physical sensor-to-siren latency under on-chip radio stress. Simulation establishes *logical* isolation—no code path makes the safety function depend on the network—but not *timing* isolation, which is a property of the silicon, the interrupt controller and the closed-source radio driver. The apparatus is specified and built (`experiments/hardware/`): a second ESP32 as an independent measuring instrument, a logic probe, and collection and analysis scripts using the same non-parametric test.

Resources needed: two ESP32 DevKit boards, a logic probe, and an authorised isolated or shielded RF environment. The attack condition would be run only on our own equipment with institutional permission; this is stated in the paper's ethics section, and no attack tooling is published in the repository.

Proposed next steps

1. Approve the hardware measurement campaign and the isolated-RF arrangement.
2. Run conditions A–C (idle, legitimate load, command stress) first; these need no RF isolation and alone support a workshop submission.
3. Add condition D (radio-layer stress) once approved. This is the headline result.
4. Target venue, for discussion: an IEEE/ACM workshop on IoT or embedded security, or *Sensors*/*Applied Sciences*, where comparable device-class work appears.

If condition D is not approved: the paper still stands on the formal isolation result plus conditions A–C, with D as future work. No claimed number depends on it.